Differential Equations Workbook Solutions

Second order equations

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SECOND ORDER LINEAR HOMOGENEOUS EQUATIONS

■ 1. Find the general solution to the homogeneous equation.

$$y'' + 9y' + 8y = 0$$

Solution:

Factor the associated characteristic equation and solve for roots.

$$r^2 + 9r + 8 = 0$$

$$(r + 8)(r + 1) = 0$$

$$r = -8, \ -1$$

These are distinct real roots, so the general solution is

$$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$y(x) = c_1 e^{-8x} + c_2 e^{-x}$$

■ 2. Find the solution to the homogeneous equation, if $y(0) = 5$ and $y'(0) = 2$.

$$y'' - 4y' + 5y = 0$$

Solution:

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The characteristic equation associated with the differential equation is

$$r^2 - 4r + 5 = 0$$

Apply the quadratic formula.

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(5)}}{2(1)}$$

$$r = \frac{4 \pm \sqrt{-4}}{2}$$

Use the imaginary number to rewrite the root of the negative number.

$$r = \frac{4 \pm \sqrt{(4)(-1)}}{2}$$

$$r = \frac{4 \pm 2\sqrt{-1}}{2}$$

$$r = 2 \pm i$$

These are complex conjugate roots. If we match up $r = \alpha \pm \beta i$ with $r = 2 \pm i$, we identify $\alpha = 2$ and $\beta = 1$, and the general solution is therefore

$$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$y(x) = e^{2x}(c_1 \cos x + c_2 \sin x)$$

The derivative of the general solution is

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$$y'(x) = 2e^{2x}(c_1 \cos x + c_2 \sin x) + e^{2x}(-c_1 \sin x + c_2 \cos x)$$

$$y'(x) = 2c_1 e^{2x} \cos x + c_2 e^{2x} \cos x + 2c_2 e^{2x} \sin x - c_1 e^{2x} \sin x$$

$$y'(x) = (2c_1 + c_2)e^{2x} \cos x + (2c_2 - c_1)e^{2x} \sin x$$

Substitute the initial conditions $y(0) = 5$ and $y'(0) = 2$ into the general solution and its derivative. We get

$$5 = e^{2(0)}(c_1 \cos(0) + c_2 \sin(0))$$

$$5 = 1(c_1(1) + 0)$$

$$c_1 = 5$$

and

$$2 = (2c_1 + c_2)e^{2(0)} \cos(0) + (2c_2 - c_1)e^{2(0)} \sin(0)$$

$$2 = (2c_1 + c_2)(1) + (2c_2 - c_1)(1)(0)$$

$$2 = 2c_1 + c_2$$

$$2 = 2(5) + c_2$$

$$c_2 = -8$$

After substituting the initial conditions, the general solution becomes

$$y(x) = e^{2x}(5 \cos x - 8 \sin x)$$

3. Find the general solution to the homogeneous equation.

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$$y'' - 10y' + 25y = 0$$

Solution:

Factor the associated characteristic equation and solve for roots.

$$r^2 - 10r + 25 = 0$$

$$(r - 5)(r - 5) = 0$$

$$r = 5, 5$$

These are equal real roots, so the general solution is

$$y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$$

$$y(x) = c_1 e^{5x} + c_2 x e^{5x}$$

■ 4. If $y(0) = 1$ and $y'(0) = 0$, find the solution to the homogeneous equation.

$$y'' + 4y' + 4y = 0$$

Solution:

Factor the associated characteristic equation and solve for roots.

$$r^2 + 4r + 4 = 0$$

$$(r + 2)(r + 2) = 0$$

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$$r = -2, \ -2$$

These are equal real roots, so the general solution is and its derivative are

$$y(x) = c_1 e^{-2x} + c_2 x e^{-2x}$$

$$y'(x) = -2c_1 e^{-2x} + c_2 e^{-2x} - 2c_2 x e^{-2x}$$

Substitute $y(0) = 1$ into $y(x)$.

$$y(0) = c_1 e^{-2(0)} + c_2(0) e^{-2(0)}$$

$$1 = c_1(1) + 0$$

$$c_1 = 1$$

Substitute $y'(0) = 0$ into $y'(x)$.

$$y'(0) = -2c_1 e^{-2(0)} + c_2 e^{-2(0)} - 2c_2(0) e^{-2(0)}$$

$$0 = -2c_1(1) + c_2(1) - 0$$

$$c_2 = 2c_1$$

$$c_2 = 2(1)$$

$$c_2 = 2$$

After substituting the initial conditions, the general solution becomes

$$y(x) = e^{-2x} + 2x e^{-2x}$$

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■ 5. If $y(0) = 2$ and $y'(0) = 3$, find the solution to the homogeneous equation.

$$y'' + 9y = 0$$

Solution:

The characteristic equation associated with the differential equation is

$$r^2 + 9 = 0$$

Apply the quadratic formula.

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{0 \pm \sqrt{0^2 - 4(1)(9)}}{2(1)}$$

$$r = \frac{\pm\sqrt{-36}}{2}$$

Use the imaginary number to rewrite the root of the negative number.

$$r = \frac{\pm\sqrt{(36)(-1)}}{2}$$

$$r = \frac{\pm 6\sqrt{-1}}{2}$$

$$r = \pm 3i$$

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These are complex conjugate roots. If we match up $r = \alpha \pm \beta i$ with $r = \pm 3i$, we identify $\alpha = 0$ and $\beta = 3$, and the general solution is therefore

$$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$y(x) = e^{(0)x}(c_1 \cos(3x) + c_2 \sin(3x))$$

$$y(x) = c_1 \cos(3x) + c_2 \sin(3x)$$

The derivative of the general solution is

$$y'(x) = -3c_1 \sin(3x) + 3c_2 \cos(3x)$$

Substitute the initial conditions $y(0) = 2$ and $y'(0) = 3$ into the general solution and its derivative. We get

$$2 = c_1 \cos(3(0)) + c_2 \sin(3(0))$$

$$2 = c_1(1) + 0$$

$$c_1 = 2$$

and

$$3 = -3c_1 \sin(3(0)) + 3c_2 \cos(3(0))$$

$$3 = 0 + 3c_2(1)$$

$$c_2 = 1$$

After substituting the initial conditions, the general solution becomes

$$y(x) = 2\cos(3x) + \sin(3x)$$

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■ 6. If $y(0) = 1$ and $y'(0) = 5$, find the solution to the homogeneous equation.

$$2y'' + 11y' + 5y = 0$$

Solution:

Factor the associated characteristic equation and solve for roots.

$$2r^2 + 11r + 5 = 0$$

$$(2r + 1)(r + 5) = 0$$

$$r = -\frac{1}{2},\ -5$$

These are distinct real roots, so the general solution is and its derivative are

$$y(x) = c_1 e^{-\frac{1}{2}x} + c_2 e^{-5x}$$

$$y'(x) = -\frac{1}{2} c_1 e^{-\frac{1}{2}x} - 5c_2 e^{-5x}$$

Substitute $y(0) = 1$ into $y(x)$.

$$y(0) = c_1 e^{-\frac{1}{2}(0)} + c_2 e^{-5(0)}$$

$$1 = c_1(1) + c_2(1)$$

$$1 = c_1 + c_2$$

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Substitute $y'(0) = 5$ into $y'(x)$.

$$y'(0) = -\frac{1}{2}c_1 e^{-\frac{1}{2}(0)} - 5c_2 e^{-5(0)}$$

$$5 = -\frac{1}{2}c_1(1) - 5c_2(1)$$

$$5 = -\frac{1}{2}c_1 - 5c_2$$

Multiply $1 = c_1 + c_2$ by 5 to get $5 = 5c_1 + 5c_2$, then add the equations.

$$5 + 5 = -\frac{1}{2}c_1 - 5c_2 + (5c_1 + 5c_2)$$

$$10 = -\frac{1}{2}c_1 + 5c_1$$

$$10 = \frac{9}{2}c_1$$

$$c_1 = \frac{20}{9}$$

Then c_2 is

$$1 = \frac{20}{9} + c_2$$

$$c_2 = -\frac{11}{9}$$

After substituting the initial conditions, the general solution becomes

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$$y(x) = \frac{20}{9}e^{-\frac{1}{2}x} - \frac{11}{9}e^{-5x}$$

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REDUCTION OF ORDER

■ 1. Use reduction of order to find the general solution to the differential equation, given $y_1 = e^{2x}$.

$$y'' + 3y' - 10y = 0$$

Solution:

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = ve^{2x}$$

$$y_2' = v'e^{2x} + 2ve^{2x}$$

$$y_2'' = v''e^{2x} + 4v'e^{2x} + 4ve^{2x}$$

Plug these values into the homogeneous differential equation.

$$v''e^{2x} + 4v'e^{2x} + 4ve^{2x} + 3(v'e^{2x} + 2ve^{2x}) - 10ve^{2x} = 0$$

$$v''e^{2x} + 7v'e^{2x} = 0$$

Make the substitution $w = v'$ and $w' = v''$.

$$w'e^{2x} + 7we^{2x} = 0$$

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$$e^{2x}\frac{dw}{dx} + 7we^{2x} = 0$$

$$\frac{dw}{dx} + 7w = 0$$

With $P(x) = 7$ and $Q(x) = 0$, the integrating factor is

$$I(x) = e^{\int P(x)\,dx}$$

$$I(x) = e^{\int 7\,dx}$$

$$I(x) = e^{7x}$$

Multiplying the first order differential equation in w by the integrating factor, and then simplifying, gives

$$\frac{d}{dx}(we^{7x}) = 0$$

Integrate both sides.

$$\int \frac{d}{dx}(we^{7x})\,dx = \int 0\,dx$$

$$we^{7x} = C$$

$$w = Ce^{-7x}$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\,dx = \int Ce^{-7x}\,dx = -\frac{1}{7}Ce^{-7x} + k$$

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We can choose any constants, so we'll choose $C = -7$ and $k = 0$ for simplicity. Then v is

$$v = e^{-7x}$$

and the second solution for the differential equation is

$$y_2 = ve^{2x}$$

$$y_2 = e^{-7x}e^{2x}$$

$$y_2 = e^{-5x}$$

Because the solutions are $y_1 = e^{2x}$ and $y_2 = e^{-5x}$, the general solution to the differential equation is

$$y(x) = c_1e^{2x} + c_2e^{-5x}$$

■ 2. Use reduction of order to find the general solution to the differential equation, given $y_1 = x^2$.

$$x^2y'' - 7xy' + 12y = 0$$

Solution:

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

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$$y_2 = vx^2$$

$$y_2' = v'x^2 + 2vx$$

$$y_2'' = v''x^2 + 4v'x + 2v$$

Plug these values into the homogeneous differential equation.

$$x^2(v''x^2 + 4v'x + 2v) - 7x(v'x^2 + 2vx) + 12vx^2 = 0$$

$$v''x^4 + 4v'x^3 + 2vx^2 - 7v'x^3 - 14vx^2 + 12vx^2 = 0$$

$$v''x^4 - 3x^3v' = 0$$

Make the substitution $w = v'$ and $w' = v''$.

$$w'x^4 - 3x^3w = 0$$

$$x^4\frac{dw}{dx} - 3x^3w = 0$$

$$\frac{dw}{dx} - \frac{3}{x}w = 0$$

With $P(x) = -3/x$ and $Q(x) = 0$, the integrating factor is

$$I(x) = e^{\int P(x)\,dx}$$

$$I(x) = e^{\int -\frac{3}{x}\,dx}$$

$$I(x) = e^{-3\ln x}$$

$$I(x) = x^{-3}$$

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Multiplying the first order differential equation in w by the integrating factor, and then simplifying, gives

$$\frac{d}{dx}(wx^{-3}) = 0$$

Integrate both sides.

$$\int \frac{d}{dx}(wx^{-3})\ dx = \int 0\ dx$$

$$wx^{-3} = C$$

$$w = Cx^3$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\ dx = \int Cx^3\ dx = \frac{C}{4}x^4 + k$$

We can choose any constants, so we'll choose $C = 4$ and $k = 0$ for simplicity. Then v is

$$v = x^4$$

and the second solution for the differential equation is

$$y_2 = vx^2$$

$$y_2 = (x^4)x^2$$

$$y_2 = x^6$$

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Because the solutions are $y_1 = x^2$ and $y_2 = x^6$, the general solution to the differential equation is

$$y(x) = c_1x^2 + c_2x^6$$

■ 3. Use reduction of order to find the general solution to the differential equation, given $y_1 = x^3$.

$$x^2y'' - 6y = x^3$$

Solution:

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = vx^3$$

$$y_2' = v'x^3 + 3vx^2$$

$$y_2'' = v''x^3 + 6v'x^2 + 6vx$$

Plug these values into the homogeneous differential equation.

$$x^2(v''x^3 + 6v'x^2 + 6vx) - 6vx^3 = x^3$$

$$v''x^5 + 6v'x^4 + 6vx^3 - 6vx^3 = = x^3$$

$$v''x^5 + 6v'x^4 = x^3$$

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$$v''x^2 + 6v'x = 1$$

Make the substitution $w = v'$ and $w' = v''$.

$$w'x^2 + 6wx = 1$$

$$w' + \frac{6}{x}w = \frac{1}{x^2}$$

With $P(x) = 6/x$ and $Q(x) = 1/x^2$, the integrating factor is

$$\mu(x) = e^{\int P(x)\,dx}$$

$$\mu(x) = e^{\int \frac{6}{x}\,dx}$$

$$\mu(x) = e^{6\ln x}$$

$$\mu(x) = x^6$$

Multiplying the first order differential equation in w by the integrating factor, and then simplifying, gives

$$\frac{d}{dx}(wx^6) = x^4$$

Integrate both sides.

$$\int \frac{d}{dx}(wx^6)\,dx = \int x^4\,dx$$

$$wx^6 = \frac{1}{5}x^5 + C$$

$$w = \frac{1}{5x} + \frac{C}{x^6}$$

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Because $w = v'$, we'll integrate w to find v.

$$v = \int w \, dx = \int \frac{1}{5x} + \frac{C}{x^6} \, dx = \frac{1}{5} \ln x + \frac{C}{5} x^{-5} + k$$

We can choose any constants, so we'll choose $C = 5$ and $k = 0$ for simplicity. Then v is

$$v = \frac{1}{5} \ln x + \frac{1}{x^5}$$

and the second solution for the differential equation is

$$y_2 = vx^3$$

$$y_2 = \left(\frac{1}{5} \ln x + \frac{1}{x^5} \right) x^3$$

$$y_2 = \frac{1}{5} x^3 \ln x + \frac{1}{x^2}$$

So the general solution to the differential equation is

$$y = c_1 x^3 + c_2 \left(\frac{1}{5} x^3 \ln x + \frac{1}{x^2} \right)$$

■ 4. Use reduction of order to find the general solution to the differential equation, given $y_1 = \cos(2x)$.

$$y'' + 4y = 0$$

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Solution:

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = v\cos(2x)$$

$$y_2' = v'\cos(2x) - 2v\sin(2x)$$

$$y_2'' = v''\cos(2x) - 4v'\sin(2x) - 4v\cos(2x)$$

Plug these values into the homogeneous differential equation.

$$v''\cos(2x) - 4v'\sin(2x) - 4v\cos(2x) + 4v\cos(2x) = 0$$

$$v''\cos(2x) - 4v'\sin(2x) = 0$$

$$v'' - 4v'\tan(2x) = 0$$

Make the substitution $w = v'$ and $w' = v''$.

$$w' - 4w\tan(2x) = 0$$

$$\frac{dw}{dx} = 4w\tan(2x)$$

$$\frac{1}{w}\,dw = 4\tan(2x)\,dx$$

Integrate both sides.

$$\int \frac{1}{w}\,dw = \int 4\tan(2x)\,dx$$

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$$\ln w = -2\ln(\cos(2x)) + C$$

$$\ln w = \ln\left(\frac{1}{\cos^2(2x)}\right) + C$$

$$w = \frac{C}{\cos^2(2x)}$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\, dx = \int \frac{C}{\cos^2(2x)}\, dx$$

$$v = \int w\, dx = \int \frac{C}{\cos^2(2x)} = \frac{C}{2}\tan(2x) + k$$

We can choose any constants, so we'll choose $C = 2$ and $k = 0$ for simplicity. Then v is

$$v = \tan(2x)$$

and the second solution for the differential equation is

$$y_2 = v\cos(2x)$$

$$y_2 = \tan(2x)\cos(2x)$$

$$y_2 = \sin(2x)$$

Because the solutions are $y_2 = \cos(2x)$ and $y_2 = \sin(2x)$, the general solution to the differential equation is

$$y(x) = c_1\cos(2x) + c_2\sin(2x)$$

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■ 5. Use reduction of order to find the general solution to the differential equation, given $y_1 = x$.

$$(x+1)y'' + xy' - y = 0$$

Solution:

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = vx$$

$$y_2' = v'x + v$$

$$y_2'' = v''x + 2v'$$

Plug these values into the homogeneous differential equation.

$$(x+1)(v''x + 2v') + x(v'x + v) - vx = 0$$

$$v''(x^2 + x) + v'(x^2 + 2x + 2) = 0$$

Make the substitution $w = v'$ and $w' = v''$.

$$w'(x^2 + x) + w(x^2 + 2x + 2) = 0$$

$$\frac{dw}{w} = -\frac{x^2 + 2x + 2}{x^2 + x}\,dx$$

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$$\frac{dw}{w} = \left(-1 - \frac{2}{x} + \frac{1}{1+x}\right)\, dx$$

$$\int \frac{dw}{w} = \int \left(-1 - \frac{2}{x} + \frac{1}{1+x}\right)\, dx$$

$$\ln w = -x - 2\ln x + \ln(x+1) + C$$

$$\ln w = -x + \ln\left(\frac{x+1}{x^2}\right) + C$$

$$w = Ce^{-x}\left(\frac{x+1}{x^2}\right)$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\, dx = \int Ce^{-x}\left(\frac{x+1}{x^2}\right)\, dx = -\frac{C}{x}e^{-x} + k$$

We can choose any constants, so we'll choose $C = 1$ and $k = 0$ for simplicity. Then v is

$$v = \frac{1}{x}e^{-x}$$

and the second solution for the differential equation is

$$y_2 = vx$$

$$y_2 = \left(\frac{1}{x}e^{-x}\right)x$$

$$y_2 = e^{-x}$$

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Because the solutions are $y_1 = x$ and $y_2 = e^{-x}$, the general solution to the differential equation is

$$y(x) = c_1 x + c_2 e^{-x}$$

■ 6. Use reduction of order to find the general solution to the differential equation, given $y_1 = e^{4x}$.

$$y'' - 5y' + 4y = 0$$

Solution:

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = ve^{4x}$$

$$y_2' = v'e^{4x} + 4ve^{4x}$$

$$y_2'' = v''e^{4x} + 8v'e^{4x} + 16ve^{4x}$$

Plug these values into the homogeneous differential equation.

$$v''e^{4x} + 8v'e^{4x} + 16ve^{4x} - 5(v'e^{4x} + 4ve^{4x}) + 4ve^{4x} = 0$$

$$v''e^{4x} + 3v'e^{4x} = 0$$

Make the substitution $w = v'$ and $w' = v''$.

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$$w'e^{4x} + 3we^{4x} = 0$$

$$\frac{dw}{dx} = -3w$$

$$\frac{dw}{w} = -3\,dx$$

$$\ln|w| = -3x$$

$$w = Ce^{-3x}$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\,dx = \int Ce^{-3x}\,dx = -\frac{1}{3}Ce^{-3x} + k$$

We can choose any constants, so we'll choose $C = -3$ and $k = 0$ for simplicity. Then v is

$$v = e^{-3x}$$

and the second solution for the differential equation is

$$y_2 = ve^{4x}$$

$$y_2 = e^{-3x}e^{4x}$$

$$y_2 = e^{x}$$

Because the solutions are $y_1 = e^{4x}$ and $y_2 = e^{x}$, the general solution to the differential equation is

$$y(x) = c_1e^{4x} + c_2e^{x}$$

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UNDETERMINED COEFFICIENTS FOR NONHOMOGENEOUS EQUATIONS

■ 1. Find the general solution to the differential equation.

$$y'' - 2y' + y = x^3 - 6x^2 + 6x + 1$$

Solution:

Solve the characteristic equation associated with the homogeneous equation,

$$y'' - 2y' + y = 0$$

$$r^2 - 2r + 1 = 0$$

$$(r-1)^2 = 0$$

$$r = 1, 1$$

to find the complementary solution with equal real roots.

$$y_c(x) = c_1 e^x + c_2 x e^x$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = Ax^3 + Bx^2 + Cx + D$$

$$y_p'(x) = 3Ax^2 + 2Bx + C$$

$$y_p''(x) = 6Ax + 2B$$

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Plug the derivatives into the original differential equation.

$$6Ax + 2B - 2(3Ax^2 + 2Bx + C) + Ax^3 + Bx^2 + Cx + D = x^3 - 6x^2 + 6x + 1$$

$$6Ax + 2B - 6Ax^2 - 4Bx - 2C + Ax^3 + Bx^2 + Cx + D = x^3 - 6x^2 + 6x + 1$$

$$Ax^3 + x^2(-6A + B) + x(6A - 4B + C) + 2B - 2C + D = x^3 - 6x^2 + 6x + 1$$

Equate coefficients.

$$A = 1$$

$$-6A + B = -6 \qquad B = 0$$

$$6A - 4B + C = 6 \qquad C = 0$$

$$2B - 2C + D = 1 \qquad D = 1$$

Then the particular solution is

$$y_p(x) = x^3 + 1$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^x + c_2 x e^x + x^3 + 1$$

■ 2. Use the method of undetermined coefficient to solve the differential equation.

$$y'' - 3y' + 2y = e^{-x}(6x^2 - 10x + 20)$$

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Solution:

Solve the characteristic equation associated with the homogeneous equation,

$$y'' - 3y' + 2y = 0$$

$$r^2 - 3r + 2 = 0$$

$$(r - 1)(r - 2) = 0$$

$$r = 1,\ 2$$

to find the complementary solution with distinct real roots.

$$y_c(x) = c_1 e^x + c_2 e^{2x}$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = e^{-x}(Ax^2 + Bx + C)$$

$$y_p'(x) = -e^{-x}(Ax^2 + Bx + C) + e^{-x}(2Ax + B) = e^{-x}(-Ax^2 + x(2A - B) + B - C)$$

$$y_p''(x) = e^{-x}(Ax^2 + x(-4A + B) + C - 2B + 2A)$$

Plug the derivatives into the original differential equation.

$$e^{-x}(Ax^2 + x(-4A + B) + C - 2B + 2A) - 3e^{-x}(-Ax^2 + x(2A - B) + B - C)$$

$$+2e^{-x}(Ax^2 + Bx + C) = e^{-x}(6x^2 - 10x + 20)$$

$$e^{-x}(6Ax^2 + x(-10A + 6B) + 2A - 5B + 6C) = e^{-x}(6x^2 - 10x + 20)$$

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Equate coefficients.

$$6A = 6 \qquad A = 1$$

$$-10A + 6B = -10 \qquad B = 0$$

$$2A - 5B + 6C = 20 \qquad C = 3$$

Then the particular solution is

$$y_p(x) = e^{-x}(x^2 + 3)$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^x + c_2 e^{2x} + e^{-x}(x^2 + 3)$$

■ 3. Find the general solution to the differential equation.

$$y'' - y' = e^{2x} + e^{3x} + e^{4x}$$

Solution:

Solve the characteristic equation associated with the homogeneous equation,

$$y'' - y' = 0$$

$$r^2 - r = 0$$

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$$r(r-1)=0$$

$$r=0,\ 1$$

to find the complementary solution with distinct real roots.

$$y_c(x)=c_1+c_2e^x$$

Our guess for the particular solution and its derivatives will be

$$y_p(x)=Ae^{2x}+Be^{3x}+Ce^{4x}$$

$$y_p'(x)=2Ae^{2x}+3Be^{3x}+4Ce^{4x}$$

$$y_p''(x)=4Ae^{2x}+9Be^{3x}+16Ce^{4x}$$

Plug the derivatives into the original differential equation.

$$4Ae^{2x}+9Be^{3x}+16Ce^{4x}-2Ae^{2x}-3Be^{3x}-4Ce^{4x}=e^{2x}+e^{3x}+e^{4x}$$

$$2Ae^{2x}+6Be^{3x}+12Ce^{4x}=e^{2x}+e^{3x}+e^{4x}$$

Equate coefficients.

$$2A=1 \qquad A=\frac{1}{2}$$

$$6B=1 \qquad B=\frac{1}{6}$$

$$12C=1 \qquad C=\frac{1}{12}$$

Then the particular solution is

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$$y_p(x) = \frac{1}{2}e^{2x} + \frac{1}{6}e^{3x} + \frac{1}{12}e^{4x}$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 + c_2e^x + \frac{1}{2}e^{2x} + \frac{1}{6}e^{3x} + \frac{1}{12}e^{4x}$$

■ 4. Solve the differential equation.

$$y'' + 9y = 12\sin(3x)$$

Solution:

Solve the characteristic equation associated with the homogeneous equation,

$$y'' + 9y = 0$$

$$r^2 + 9 = 0$$

$$r = \pm 3i$$

to find the complementary solution with complex conjugate roots.

$$y_c(x) = c_1\sin(3x) + c_2\cos(3x)$$

Our guess for the particular solution and its derivatives will be

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$$y_p(x) = Ax\sin(3x) + Bx\cos(3x)$$

$$y_p'(x) = A\sin(3x) + 3Ax\cos(3x) + B\cos(3x) - 3Bx\sin(3x)$$

$$y_p''(x) = 6A\cos(3x) - 9Ax\sin(3x) - 6B\sin(3x) - 9Bx\cos(3x)$$

Plug the derivatives into the original differential equation.

$$6A\cos(3x) - 9Ax\sin(3x) - 6B\sin(3x) - 9Bx\cos(3x)$$

$$+9(Ax\sin(3x) + Bx\cos(3x)) = 12\sin(3x)$$

$$6A\cos(3x) - 9Ax\sin(3x) - 6B\sin(3x) - 9Bx\cos(3x)$$

$$+9Ax\sin(3x) + 9Bx\cos(3x) = 12\sin(3x)$$

$$6A\cos(3x) - 6B\sin(3x) = 12\sin(3x)$$

Equate coefficients.

$$6A = 0 \qquad A = 0$$

$$-6B = 12 \qquad B = -2$$

Then the particular solution is

$$y_p(x) = -2x\cos(3x)$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y_c(x) = c_1\sin(3x) + c_2\cos(3x) - 2x\cos(3x)$$

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■ 5. Solve the differential equation.

$$y'' - 4y' + 4y = e^{2x}$$

Solution:

Solve the characteristic equation associated with the homogeneous equation,

$$y'' - 4y' + 4y = 0$$

$$r^2 - 4r + 4 = 0$$

$$(r-2)^2 = 0$$

$$r = 2, 2$$

to find the complementary solution with equal real roots.

$$y_c(x) = c_1 e^{2x} + c_2 x e^{2x}$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = Ax^2 e^{2x}$$

$$y_p'(x) = 2Ax^2 e^{2x} + 2Axe^{2x} = 2Ae^{2x}(x^2 + x)$$

$$y_p''(x) = 4Ae^{2x}(x^2 + x) + 2Ae^{2x}(2x + 1) = 2Ae^{2x}(2x^2 + 4x + 1)$$

Plug the derivatives into the original differential equation.

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$$2Ae^{2x}(2x^2+4x+1)-4(2Ae^{2x}(x^2+x))+4Ax^2e^{2x}=e^{2x}$$

$$2Ae^{2x}(2x^2+4x+1)-8Ae^{2x}(x^2+x))+4Ax^2e^{2x}=e^{2x}$$

Equate coefficients.

$$2Ae^{2x}=e^{2x} \qquad A=\frac{1}{2}$$

Then the particular solution is

$$y_p(x)=\frac{1}{2}x^2e^{2x}$$

and the general solution is

$$y(x)=c_1e^{2x}+c_2xe^{2x}+\frac{1}{2}x^2e^{2x}$$

6. Solve the differential equation.

$$y''-4y'+3y=e^x+10\sin x+3x$$

Solution:

Solve the characteristic equation associated with the homogeneous equation,

$$y''-4y'+3y=0$$

$$r^2-4r+3=0$$

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$$(r-1)(r-3)=0$$

$$r=1,\ 3$$

to find the complementary solution with distinct real roots.

$$y_c(x)=c_1e^x+c_2e^{3x}$$

Our guess for the particular solution and its derivatives will be

$$y_p(x)=Axe^x+B\sin x+C\cos x+Dx+E$$

$$y_p'(x)=Ae^x+Axe^x+B\cos x-C\sin x+D$$

$$y_p''(x)=2Ae^x+Axe^x-B\sin x-C\cos x$$

Plug the derivatives into the original differential equation.

$$2Ae^x+Axe^x-B\sin x-C\cos x-4(Ae^x+Axe^x+B\cos x-C\sin x+D)+$$

$$+3(Axe^x+B\sin x+C\cos x+Dx+E)=e^x+10\sin x+3x$$

$$-2Ae^x+\sin x(4C+2B)+\cos x(2C-4B)+3Dx+3E-4D=e^x+10\sin x+3x$$

Equate coefficients.

$$-2A=1 \qquad A=-1/2$$

$$4C-2B=10 \qquad B=1$$

$$2C-4B=0 \qquad C=2$$

$$3D=3 \qquad D=1$$

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$$3E - 4D = 0 \qquad E = \frac{4}{3}$$

Then the particular solution is

$$y_p(x) = -\frac{1}{2}xe^x + \sin x + 2\cos x + x + \frac{4}{3}$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1e^x + c_2e^{3x} - \frac{1}{2}xe^x + \sin x + 2\cos x + x + \frac{4}{3}$$

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VARIATION OF PARAMETERS FOR NONHOMOGENEOUS EQUATIONS

■ 1. Find the general solution to the differential equation.

$$y'' - 10y' + 25y = x^3 e^{5x}$$

Solution:

Solve the associated homogeneous equation,

$$y'' - 10y' + 25y = 0$$

$$r^2 - 10r + 25 = 0$$

$$(r - 5)(r - 5) = 0$$

$$r = 5, 5$$

to find equal real roots, then build the complementary solution.

$$y_c(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$$

$$y_c(x) = c_1 e^{5x} + c_2 x e^{5x}$$

The fundamental set of solutions and its derivatives are therefore

$$\{y_1, y_2\} = \{e^{5x}, xe^{5x}\}$$

$$\{y_1', y_2'\} = \{5e^{5x}, e^{5x} + 5xe^{5x}\}$$

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Create a system of linear equations,

$$u_1'y_1 + u_2'y_2 = 0 \qquad u_1'y_1' + u_2'y_2' = g(x)$$

$$u_1'e^{5x} + u_2'xe^{5x} = 0 \qquad u_1'(5e^{5x}) + u_2'(e^{5x} + 5xe^{5x}) = x^3e^{5x}$$

$$5u_1'e^{5x} + u_2'e^{5x} + 5u_2'xe^{5x} = x^3e^{5x}$$

then solve it for u_2',

$$5u_1'e^{5x} + 5u_2'xe^{5x} - (5u_1'e^{5x} + u_2'e^{5x} + 5u_2'xe^{5x}) = 0 - x^3e^{5x}$$

$$5u_1'e^{5x} + 5u_2'xe^{5x} - 5u_1'e^{5x} - u_2'e^{5x} - 5u_2'xe^{5x} = -x^3e^{5x}$$

$$-u_2'e^{5x} = -x^3e^{5x}$$

$$u_2' = x^3$$

and then u_1'.

$$u_1'e^{5x} + (x^3)xe^{5x} = 0$$

$$u_1'e^{5x} = -x^4e^{5x}$$

$$u_1' = -x^4$$

Then u_1 and u_2 are

$$u_1 = \int u_1' = \int -x^4\,dx = -\frac{1}{5}x^5$$

$$u_2 = \int u_2' = \int x^3\,dx = \frac{1}{4}x^4$$

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The particular solution is

$$y_p(x) = u_1 y_1 + u_2 y_2$$

$$y_p(x) = -\frac{1}{5}x^5(e^{5x}) + \frac{1}{4}x^4(xe^{5x})$$

$$y_p(x) = -\frac{1}{5}x^5 e^{5x} + \frac{1}{4}x^5 e^{5x}$$

$$y_p(x) = \frac{x^5 e^{5x}}{20}$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1 e^{5x} + c_2 x e^{5x} + \frac{x^5 e^{5x}}{20}$$

■ 2. Find the general solution to the differential equation.

$$y'' - 6y' + 8y = e^{3x} + 5$$

Solution:

Solve the associated homogeneous equation,

$$y'' - 6y' + 8y = 0$$

$$r^2 - 6r + 8 = 0$$

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$$(r-2)(r-4)=0$$

$$r=2,\,4$$

to find distinct real roots, then build the complementary solution.

$$y_c(x)=c_1e^{r_1x}+c_2e^{r_2x}$$

$$y_c(x)=c_1e^{2x}+c_2e^{4x}$$

The fundamental set of solutions and its derivatives are therefore

$$\{y_1,y_2\}=\{e^{2x},\,e^{4x}\}$$

$$\{y_1',y_2'\}=\{2e^{2x},\,4e^{4x}\}$$

Create a system of linear equations,

$$u_1'y_1+u_2'y_2=0$$

$$u_1'e^{2x}+u_2'e^{4x}=0$$

$$u_1'y_1'+u_2'y_2'=g(x)$$

$$u_1'(2e^{2x})+u_2'(4e^{4x})=e^{3x}+5$$

$$2u_1'e^{2x}+4u_2'e^{4x}=e^{3x}+5$$

then solve it for u_2',

$$2u_1'e^{2x}+4u_2'e^{4x}-2(u_1'e^{2x}+u_2'e^{4x})=e^{3x}+5-0$$

$$2u_2'e^{4x}=e^{3x}+5$$

$$u_2'=\frac{e^{3x}+5}{2e^{4x}}$$

and then u_1'.

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$$u_1'e^{2x} + u_2'e^{4x} = 0$$

$$u_1'e^{2x} + \frac{e^{3x}+5}{2e^{4x}} \cdot e^{4x} = 0$$

$$u_1'e^{2x} + \frac{e^{3x}+5}{2} = 0$$

$$u_1' = -\frac{e^{3x}+5}{2e^{2x}}$$

Then u_1 and u_2 are

$$u_1 = \int u_1' = \int -\frac{e^{3x}+5}{2e^{2x}}\,dx = -\frac{e^x}{2} + \frac{5}{4}e^{-2x}$$

$$u_2 = \int u_2' = \int \frac{e^{3x}+5}{2e^{4x}}\,dx = -\frac{1}{2}e^{-x} - \frac{5}{8}e^{-4x}$$

The particular solution is

$$y_p(x) = u_1y_1 + u_2y_2$$

$$y_p(x) = \left(-\frac{e^x}{2} + \frac{5}{4}e^{-2x}\right)(e^{2x}) + \left(-\frac{1}{2}e^{-x} - \frac{5}{8}e^{-4x}\right)(e^{4x})$$

$$y_p(x) = -e^{3x} + \frac{5}{8}$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1e^{2x} + c_2e^{4x} - e^{3x} + \frac{5}{8}$$

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■ 3. Find the general solution to the differential equation.

$$y'' - 7y' + 12y = \frac{e^{5x}}{e^{2x} + 1}$$

Solution:

Solve the associated homogeneous equation,

$$y'' - 7y' + 12y = 0$$

$$r^2 - 7r + 12 = 0$$

$$(r - 3)(r - 4) = 0$$

$$r = 3, 4$$

to find distinct real roots, then build the complementary solution.

$$y_c(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$y_c(x) = c_1 e^{3x} + c_2 e^{4x}$$

The fundamental set of solutions and its derivatives are therefore

$$\{y_1, y_2\} = \{e^{3x}, e^{4x}\}$$

$$\{y_1', y_2'\} = \{3e^{3x}, 4e^{4x}\}$$

Create a system of linear equations,

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$$u_1'y_1 + u_2'y_2 = 0$$

$$u_1'y_1' + u_2'y_2' = g(x)$$

$$u_1'e^{3x} + u_2'e^{4x} = 0$$

$$u_1'(3e^{3x}) + u_2'(4e^{4x}) = \frac{e^{5x}}{e^{2x}+1}$$

$$3u_1'e^{3x} + 4u_2'e^{4x} = \frac{e^{5x}}{e^{2x}+1}$$

then solve it for u_2',

$$u_1' = -u_2'e^{x}$$

$$u_2'e^{4x} = \frac{e^{5x}}{e^{2x}+1}$$

$$u_2' = \frac{e^{x}}{e^{2x}+1}$$

and then u_1'.

$$u_1' = -\frac{e^{2x}}{e^{2x}+1}$$

Then using the substitution $t = e^{2x}$ and $dt = 2e^2x\ dx$ to find u_1, and the substitution $t = e^{x}$ and $dt = e^{x}\ dx$ to find u_2,

$$u_1 = \int u_1' = \int -\frac{e^{2x}}{e^{2x}+1}\ dx = -\frac{1}{2}\ln(e^{2x}+1)$$

$$u_2 = \int u_2' = \int \frac{e^{x}}{e^{2x}+1}\ dx = \arctan e^{x}$$

the particular solution is

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$$y_p(x) = u_1y_1 + u_2y_2$$

$$y_p(x) = -\frac{1}{2}\ln(e^{2x} + 1)(e^{3x}) + \arctan e^x(e^{4x})$$

$$y_p(x) = -\frac{e^{3x}}{2}\ln(e^{2x} + 1) + e^{4x}\arctan e^x$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1e^{3x} + c_2e^{4x} - \frac{e^{3x}}{2}\ln(e^{2x} + 1) + e^{4x}\arctan e^x$$

■ 4. Find the general solution to the differential equation.

$$y'' - 6y' + 9y = e^{3x}\sin(2x + 1)$$

Solution:

Solve the associated homogeneous equation,

$$y'' - 6y' + 9y = 0$$

$$r^2 - 6r + 9 = 0$$

$$(r - 3)^2 = 0$$

$$r = 3, 3$$

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to find equal real roots, then build the complementary solution.

$$y_c(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$$

$$y_c(x) = c_1 e^{3x} + c_2 x e^{3x}$$

The fundamental set of solutions and its derivatives are therefore

$$\{y_1, y_2\} = \{e^{3x},\ xe^{3x}\}$$

$$\{y_1', y_2'\} = \{3e^{3x},\ e^{3x} + 3xe^{3x}\}$$

Create a system of linear equations,

$$u_1' y_1 + u_2' y_2 = 0 \qquad u_1' y_1' + u_2' y_2' = g(x)$$

$$u_1' e^{3x} + u_2' x e^{3x} = 0 \qquad u_1'(3e^{3x}) + u_2'(e^{3x} + 3xe^{3x}) = e^{3x}\sin(2x+1)$$

$$3u_1' e^{3x} + u_2' e^{3x} + 3u_2' x e^{3x} = e^{3x}\sin(2x+1)$$

then solve it for u_2'

$$3u_1' e^{3x} + 3u_2' x e^{3x} - (3u_1' e^{3x} + u_2' e^{3x} + 3u_2' x e^{3x}) = 0 - e^{3x}\sin(2x+1)$$

$$-u_2' e^{3x} = -\,e^{3x}\sin(2x+1)$$

$$u_2' = \sin(2x+1)$$

and then hen u_1'.

$$u_1' e^{3x} + u_2' x e^{3x} = 0$$

$$u_1' = -\,x u_2'$$

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$$u_1' = -x\sin(2x+1)$$

Then using integration by parts with $t = -x$, $dt = -dx$, $dv = \sin(2x+1)\,dx$, and $v = -(1/2)\cos(2x+1)$ to find u_1, and integrating directly to find u_2,

$$u_1 = \int u_1' = \int -x\sin(2x+1)\,dx = \frac{1}{2}x\cos(2x+1) - \frac{1}{4}\sin(2x+1)$$

$$u_2 = \int u_2' = \int \sin(2x+1)\,dx = -\frac{1}{2}\cos(2x+1)$$

the particular solution is

$$y_p(x) = u_1y_1 + u_2y_2$$

$$y_p(x) = \left(\frac{1}{2}x\cos(2x+1) - \frac{1}{4}\sin(2x+1)\right)(e^{3x}) - \frac{1}{2}\cos(2x+1)(xe^{3x})$$

$$y_p(x) = \frac{1}{2}xe^{3x}\cos(2x+1) - \frac{1}{4}e^{3x}\sin(2x+1) - \frac{1}{2}xe^{3x}\cos(2x+1)$$

$$y_p(x) = -\frac{1}{4}e^{3x}\sin(2x+1)$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1e^{3x} + c_2xe^{3x} - \frac{1}{4}e^{3x}\sin(2x+1)$$

■ 5. Find the general solution to the differential equation.

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$$2y'' + 18y = \sin^4(3x)$$

Solution:

Rewrite the equation in standard form.

$$y'' + 9y = \frac{\sin^4(3x)}{2}$$

Solve the associated homogeneous equation,

$$y'' + 9y = 0$$

$$r^2 + 9 = 0$$

$$r = \pm 3i$$

to find complex conjugate roots, then build the complementary solution.

$$y_c(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$y_c(x) = c_1 \cos(3x) + c_2 \sin(3x)$$

The fundamental set of solutions and its derivatives are therefore

$$\{y_1, y_2\} = \{\cos(3x),\ \sin(3x)\}$$

$$\{y_1', y_2'\} = \{-3\sin(3x),\ 3\cos(3x)\}$$

Create a system of linear equations,

$$u_1'y_1 + u_2'y_2 = 0 \qquad u_1'y_1' + u_2'y_2' = g(x)$$

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$$u_1' \cos(3x) + u_2' \sin(3x) = 0$$

$$u_1'(-3\sin(3x)) + u_2'(3\cos(3x)) = \frac{\sin^4(3x)}{2}$$

$$-3u_1' \sin(3x) + 3u_2' \cos(3x) = \frac{\sin^4(3x)}{2}$$

then solve it for u_2',

$$u_1' = -u_2' \frac{\sin(3x)}{\cos(3x)}$$

$$-3\left(-u_2' \frac{\sin(3x)}{\cos(3x)}\right)\sin(3x) + 3u_2' \cos(3x) = \frac{\sin^4(3x)}{2}$$

$$u_2' = \frac{\cos(3x)\sin^4(3x)}{6}$$

and then u_1'.

$$u_1' = -u_2' \frac{\sin(3x)}{\cos(3x)}$$

$$u_1' = -\frac{\cos(3x)\sin^4(3x)}{6} \cdot \frac{\sin(3x)}{\cos(3x)}$$

$$u_1' = -\frac{\sin^5(3x)}{6}$$

Then using the substitution $t = \cos(3x)$ and $dt = -(1/3)\sin(3x)\ dx$ to find u_1, and the substitution $t = \sin(3x)$ and $dt = (1/3)\cos(3x)\ dx$ to find u_2,

$$u_1 = \int u_1' = \int -\frac{\sin^5(3x)}{6}\ dx = -\frac{1}{6}\int (1-\cos^2(3x))^2 \sin(3x)\ dx$$

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$$= \frac{1}{18}\left(\cos(3x) - \frac{2}{3}\cos^3(3x) + \frac{1}{5}\cos^5(3x)\right)$$

$$u_2 = \int u_2' = \int \frac{\cos(3x)\sin^4(3x)}{6}\,dx = \frac{t^5}{90} = \frac{\sin^5(3x)}{90}$$

the particular solution is

$$y_p(x) = u_1 y_1 + u_2 y_2$$

$$y_p(x) = \frac{1}{18}\left(\cos(3x) - \frac{2}{3}\cos^3(3x) + \frac{1}{5}\cos^5(3x)\right)(\cos(3x)) + \frac{\sin^5(3x)}{90}(\sin(3x))$$

$$y_p(x) = \frac{1}{18}\left(\cos(3x) - \frac{2}{3}\cos^3(3x) + \frac{1}{5}\cos^5(3x)\right)(\cos(3x)) + \frac{\sin^5(3x)}{90}(\sin(3x))$$

$$y_p(x) = \frac{1}{18}\cos^2(3x) - \frac{1}{27}\cos^4(3x) + \frac{1}{90}\cos^6(3x) + \frac{1}{90}\sin^6(3x)$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1\cos(3x) + c_2\sin(3x) + \frac{1}{18}\cos^2(3x) - \frac{1}{27}\cos^4(3x) + \frac{1}{90}\cos^6(3x) + \frac{1}{90}\sin^6(3x)$$

6. Find the general solution to the differential equation.

$$y'' - 4y' + 5y = e^{2x}\ln(1 + \sin x)$$

Solution:

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Solve the associated homogeneous equation,

$$y'' - 4y' + 5y = 0$$

$$r^2 - 4r + 5 = 0$$

$$r = 2 \pm i$$

to find complex conjugate roots, then build the complementary solution.

$$y_c(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$y_c(x) = e^{2x}(c_1 \cos x + c_2 \sin x)$$

The fundamental set of solutions and its derivatives are therefore

$$\{y_1, y_2\} = \{e^{2x} \cos x,\ e^{2x} \sin x\}$$

$$\{y_1', y_2'\} = \{2e^{2x} \cos x - e^{2x} \sin x,\ 2e^{2x} \sin x + e^{2x} \cos x\}$$

Create a system of linear equations,

$$u_1' y_1 + u_2' y_2 = 0$$

$$u_1' e^{2x} \cos x + u_2' e^{2x} \sin x = 0$$

and

$$u_1' y_1' + u_2' y_2' = g(x)$$

$$u_1'(2e^{2x} \cos x - e^{2x} \sin x) + u_2'(2e^{2x} \sin x + e^{2x} \cos x) = e^{2x} \ln(1 + \sin x)$$

then solve it for u_2' by rewriting the first equation for u_1',

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$$u_1' e^{2x} \cos x = -u_2' e^{2x} \sin x$$

$$u_1' = -u_2' \tan x$$

and then plugging this value into the second equation in the system.

$$-u_2' \tan x(2e^{2x} \cos x - e^{2x} \sin x) + u_2'(2e^{2x} \sin x + e^{2x} \cos x) = e^{2x} \ln(1 + \sin x)$$

$$u_2'(2e^{2x} \sin x + e^{2x} \cos x) - u_2'(2e^{2x} \sin x - e^{2x} \sin x \tan x) = e^{2x} \ln(1 + \sin x)$$

$$u_2'(2e^{2x} \sin x + e^{2x} \cos x - 2e^{2x} \sin x + e^{2x} \sin x \tan x) = e^{2x} \ln(1 + \sin x)$$

$$u_2'(e^{2x} \cos x + e^{2x} \sin x \tan x) = e^{2x} \ln(1 + \sin x)$$

$$u_2'(\cos x + \sin x \tan x) = \ln(1 + \sin x)$$

$$u_2' = \frac{\ln(1 + \sin x)}{\cos x + \frac{\sin^2 x}{\cos x}}$$

$$u_2' = \frac{\ln(1 + \sin x)}{\frac{\cos^2 x + \sin^2 x}{\cos x}}$$

$$u_2' = \frac{\ln(1 + \sin x)\cos x}{\cos^2 x + \sin^2 x}$$

$$u_2' = \ln(1 + \sin x)\cos x$$

Then find u_1'.

$$u_1' = -\ln(1 + \sin x)\cos x \tan x$$

$$u_1' = -\ln(1 + \sin x)\sin x$$

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$$u_1' = -\sin x \ln(1 + \sin x)$$

Then using integration by parts with $u = \ln(1 + \sin x)$, $du = \cos x/(1 + \sin x)\ dx$, $dv = -\sin x\ dx$, and $v = \cos x$ to find u_1, and using a substitution with $t = \sin x$ and $dt = \cos x\ dx$, followed by integration by parts with $u = \ln(1 + t)\ du$, $du = 1/(t + 1)\ dt$, $dv = dt$, and $v = t$ to find u_2,

$$u_1 = \int u_1' = \int -\sin x \ln(1 + \sin x)\ dx = \cos x \ln(1 + \sin x) - x - \cos x$$

$$u_2 = \int u_2' = \int \cos x \ln(1 + \sin x)\ dx = (\sin x + 1)\ln(\sin x + 1) - \sin x$$

the particular solution is

$$y_p(x) = u_1 y_1 + u_2 y_2$$

$$y_p(x) = (\cos x \ln(1 + \sin x) - x - \cos x)(e^{2x} \cos x)$$

$$+((\sin x + 1)\ln(\sin x + 1) - \sin x)(e^{2x} \sin x)$$

$$y_p(x) = e^{2x}((\sin x + 1)\ln(\sin x + 1) - x \cos x - 1)$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y = e^{2x}(c_1 \cos x + c_2 \sin x + (\sin x + 1)\ln(\sin x + 1) - x \cos x - 1)$$

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FUNDAMENTAL SOLUTION SETS AND THE WRONSKIAN

■ 1. Use the Wronskian to determine whether $\{y_1, y_2\} = \{e^x, e^{x+1}\}$ is the fundamental set of solutions for the differential equation.

$$y'' = y$$

Solution:

First we verify that y_1 and y_2 are solutions of the differential equation.

$$y_1'' = (e^x)'' = (e^x)' = e^x$$

$$y'' = y$$

$$e^x = e^x$$

$$y_2'' = (e^{x+1})'' = (e^{x+1})' = e^{x+1}$$

$$y'' = y$$

$$e^{x+1} = e^{x+1}$$

So $\{y_1, y_2\}$ is a set of solutions for differential equation. To verify whether the solution set is a fundamental set, we'll take the Wronskian.

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

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$$W(e^x, e^{x+1}) = \begin{vmatrix} e^x & e^{x+1} \\ e^x & e^{x+1} \end{vmatrix}$$

$$W(e^x, e^{x+1}) = (e^x)(e^{x+1}) - (e^{x+1})(e^x)$$

$$W(e^x, e^{x+1}) = 0$$

Since the Wronskian is zero, $\{y_1, y_2\}$ is not a fundamental set of solutions.

■ 2. Determine whether $\{y_1, y_2\} = \{\cos(2x),\ \cos(2x+1)\}$ is a fundamental set of solutions for the differential equation.

$$y'' + 4y = 0$$

Solution:

First we verify that y_1 and y_2 are solutions of the differential equation.

$$y_1'' = (\cos(2x))'' = (-2\sin(2x))' = -4\cos(2x)$$

$$y'' + 4y = 0$$

$$-4\cos(2x) + 4\cos(2x) = 0$$

$$0 = 0$$

$$y_2'' = (\cos(2x+1))'' = (-2\sin(2x+1))' = -4\cos(2x+1)$$

$$y'' + 4y = 0$$

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$$-4\cos(2x+1)+4\cos(2x+1)=0$$

$$0=0$$

So $\{y_1, y_2\}$ is a set of solutions for differential equation. To verify whether the solution set is a fundamental set, we'll take the Wronskian.

$$W(y_1,y_2)=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W(\cos(2x),\cos(2x+1))=\begin{vmatrix} \cos(2x) & \cos(2x+1) \\ -2\sin(2x) & -2\sin(2x+1) \end{vmatrix}$$

$$W(\cos(2x),\cos(2x+1))=(\cos(2x)(-2\sin(2x+1))-(\cos(2x+1))(-2\sin(2x))$$

$$W(\cos(2x),\cos(2x+1))=-2\sin 1$$

Since the Wronskian is non-zero, $\{y_1, y_2\}$ is a fundamental set of solutions.

3. Find the Wronskian for the solution set.

$$\{e^{3x},\ xe^{3x}\}$$

Solution:

Find the derivatives of the solutions in the set.

$$(e^{3x})'=3e^{3x}$$

$$(xe^{3x})'=e^{3x}+3xe^{3x}$$

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So the Wronskian for the set is

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W(e^{3x}, xe^{3x}) = \begin{vmatrix} e^{3x} & xe^{3x} \\ 3e^{3x} & e^{3x} + 3xe^{3x} \end{vmatrix}$$

$$W(e^{3x}, xe^{3x}) = e^{3x}(e^{3x} + 3xe^{3x}) - xe^{3x}(3e^{3x})$$

$$W(e^{3x}, xe^{3x}) = e^{6x} + 3xe^{6x} - 3xe^{6x}$$

$$W(e^{3x}, xe^{3x}) = e^{6x}$$

4. Find the Wronskian for the solution set.

$$\{e^{2x}\cos(2x),\ e^{2x}\sin(2x)\}$$

Solution:

Find the derivatives of the solutions in the set.

$$(e^{2x}\cos(2x))' = 2e^{2x}\cos(2x) - 2e^{2x}\sin(2x)$$

$$(e^{2x}\sin(2x))' = 2e^{2x}\sin(2x) + 2e^{2x}\cos(2x)$$

So the Wronskian for the set is

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1'$$

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$$W(e^{2x}\cos(2x), e^{2x}\sin(2x)) = \begin{vmatrix} e^{2x}\cos(2x) & e^{2x}\sin(2x) \\ 2e^{2x}\cos(2x) - 2e^{2x}\sin(2x) & 2e^{2x}\sin(2x) + 2e^{2x}\cos(2x) \end{vmatrix}$$

$$W(e^{2x}\cos(2x), e^{2x}\sin(2x)) = e^{2x}\cos(2x)(2e^{2x}\cos(2x) - 2e^{2x}\sin(2x))$$

$$-e^{2x}\sin(2x)(2e^{2x}\sin(2x) + 2e^{2x}\cos(2x))$$

$$W(e^{2x}\cos(2x), e^{2x}\sin(2x)) = 2e^{4x}$$

■ 5. Find the fundamental set of solutions for the the second order differential equation.

$$y'' + 2y' + y = 0$$

Solution:

Normally, we would find the general solution to this second order equation by solving the associated characteristic equation,

$$r^2 + 2r + 1 = 0$$

$$(r+1)^2 = 0$$

$$r = -1, \ -1$$

to get the complementary solution.

$$y_c(x) = c_1 e^{-x} + c_2 x e^{-x}$$

The derivative of the complementary solution is

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$$y_c'(x) = -c_1e^{-x} + c_2e^{-x} - c_2xe^{-x}$$

If we now plug in the initial conditions $y(x_0) = 1$ and $y'(x_0) = 0$, and $y(x_0) = 0$ and $y'(x_0) = 1$, we'll be able to generate another fundamental set of solutions. We'll use $x_0 = 0$, since we can choose any value of x_0 that we like, and using $x_0 = 0$ will be the easiest.

For $y(0) = 1$, $\quad 1 = c_1 \cdot 1 + c_2 \cdot 0$

For $y'(0) = 0$, $\quad 0 = c_1 - c_2 + 0$

Then this system gives $c_1 = 1$ and $c_2 = 1$, so we could say that one solution in our fundamental set of solutions is

$$y_1(x) = e^{-x} + xe^{-x}$$

The second set of initial conditions gives

For $y(0) = 0$, $\quad 0 = c_1 + 0$

For $y'(0) = 1$, $\quad 1 = -c_1 + c_2$

Then this system gives $c_1 = 0$ and $c_2 = 1$, so we could say that the other solution in our fundamental set is

$$y_2(x) = xe^{-x}$$

So one fundamental set of solutions is $\{y_1, y_2\} = \{e^{-x} + xe^{-x}, xe^{-x}\}$.

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■ 6. Find the fundamental set of solutions for the the second order differential equation, generated by solutions of the initial value problems with $y(0) = 1$ and $y'(0) = 0$, and $y(0) = 0$ and $y'(0) = 1$.

$$y'' + 4y' = 0$$

Solution:

Normally, we would find the general solution to this second order equation by solving the associated characteristic equation,

$$r^2 + 4r = 0$$

$$r = 0, \ -4$$

to get the complementary solution.

$$y_c(x) = c_1 + c_2 e^{-4x}$$

The derivative of the complementary solution is

$$y_c'(x) = -4c_2 e^{-4x}$$

If we now plug in the initial conditions $y(x_0) = 1$ and $y'(x_0) = 0$, and $y(x_0) = 0$ and $y'(x_0) = 1$, we'll be able to generate another fundamental set of solutions. We'll use $x_0 = 0$, since we can choose any value of x_0 that we like, and using $x_0 = 0$ will be the easiest.

For $y(0) = 1$, $\quad 1 = c_1 + c_2$

For $y'(0) = 0$, $\quad 0 = -4c_2$

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Then this system gives $c_1 = 1$ and $c_2 = 0$, so we could say that one solution in our fundamental set of solutions is

$$y_1(x) = 1$$

The second set of initial conditions gives

For $y(0) = 0$, $\qquad 0 = c_1 + c_2$

For $y'(0) = 1$, $\qquad 1 = -4c_2$

Then this system gives $c_1 = 1/4$ and $c_2 = -1/4$, so we could say that the other solution in our fundamental set is

$$y_2(x) = \frac{1}{4} - \frac{1}{4}e^{-4x}$$

So one fundamental set of solutions is

$$\{y_1, y_2\} = \left\{1, \frac{1}{4} - \frac{1}{4}e^{-4x}\right\}$$

We can verify that this set is a fundamental set by taking the Wronskian.

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W\left(1, \frac{1}{4} - \frac{1}{4}e^{-4x}\right) = \begin{vmatrix} 1 & \frac{1}{4} - \frac{1}{4}e^{-4x} \\ 0 & e^{-4x} \end{vmatrix}$$

$$W\left(1, \frac{1}{4} - \frac{1}{4}e^{-4x}\right) = e^{-4x}$$

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VARIATION OF PARAMETERS WITH THE WRONSKIAN

■ 1. Solve the differential equation.

$$y'' + 7y' + 12y = e^x + 7e^{3x}$$

Solution:

The associated homogeneous equation is

$$y'' + 7y' + 12y = 0$$

and solving the characteristic equation gives distinct real roots.

$$r^2 - 7r + 12 = 0$$

$$(r + 3)(r + 4) = 0$$

$$r = -3, \ -4$$

With distinct real roots, the complementary solution is

$$y_c(x) = c_1 e^{-3x} + c_2 e^{-4x}$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{e^{-3x}, \ e^{-4x}\}$$

Find the Wronskian for the fundamental solution set.

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$$W(e^{-3x}, e^{-4x}) = \begin{vmatrix} e^{-3x} & e^{-4x} \\ -3e^{-3x} & -4e^{-4x} \end{vmatrix}$$

$$W(e^{-3x}, e^{-4x}) = -4e^{-7x} + 3e^{-7x}$$

$$W(e^{-3x}, e^{-4x}) = -e^{-7x}$$

Then the particular solution is

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -e^{-3x} \int \frac{e^{-4x}(e^x + 7e^{3x})}{-e^{-7x}}\, dx + e^{-4x} \int \frac{e^{-3x}(e^x + 7e^{3x})}{-e^{-7x}}\, dx$$

$$y_p(x) = -e^{-3x} \int (-e^{4x} - 7e^{6x})\, dx + e^{-4x} \int (-e^{5x} - 7e^{7x})\, dx$$

$$y_p(x) = -e^{-3x} \left(-\frac{1}{4}e^{4x} - \frac{7}{6}e^{6x} \right) + e^{-4x} \left(-\frac{1}{5}e^{5x} - e^{7x} \right)$$

$$y_p(x) = \frac{1}{20}e^x + \frac{1}{6}e^{3x}$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^{-3x} + c_2 e^{-4x} + \frac{1}{20}e^x + \frac{1}{6}e^{3x}$$

2. Solve the differential equation.

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$$y'' - 2y' = \frac{e^{3x}}{e^{2x} + 1}$$

Solution:

The associated homogeneous equation is

$$y'' - 2y' = 0$$

and solving the characteristic equation gives distinct real roots.

$$r^2 - 2r = 0$$

$$r(r - 2) = 0$$

$$r = 0,\ 2$$

With distinct real roots, the complementary solution is

$$y_c(x) = c_1 + c_2 e^{2x}$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{1,\ e^{2x}\}$$

Find the Wronskian for the fundamental solution set.

$$W(1, e^{2x}) = \begin{vmatrix} 1 & e^{2x} \\ 0 & 2e^{2x} \end{vmatrix}$$

$$W(1, e^{2x}) = 2e^{2x}$$

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Then the particular solution is

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -\int \frac{e^{2x}(e^{3x})}{2e^{2x}(e^{2x}+1)}\, dx + e^{2x} \int \frac{e^{3x}}{2e^{2x}(e^{2x}+1)}\, dx$$

$$y_p(x) = -\int \frac{e^{3x}}{2(e^{2x}+1)}\, dx + e^{2x} \int \frac{e^{x}}{2(e^{2x}+1)}\, dx$$

$$y_p(x) = -\left(\frac{1}{2}e^x - \frac{1}{2}\arctan(e^x)\right) + \frac{1}{2}e^{2x}\arctan(e^x)$$

$$y_p(x) = -\frac{1}{2}e^x + \frac{1}{2}\arctan(e^x) + \frac{1}{2}e^{2x}\arctan(e^x)$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 + c_2 e^{2x} - \frac{1}{2}e^x + \frac{1}{2}\arctan(e^x) + \frac{1}{2}e^{2x}\arctan(e^x)$$

3. Solve the differential equation.

$$y'' + 8y' + 16y = 21x^5 e^{-4x}$$

Solution:

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The associated homogeneous equation is

$$y'' + 8y' + 16y = 0$$

and solving the characteristic equation gives equal real roots.

$$r^2 + 8r + 16 = 0$$

$$(r + 4)^2 = 0$$

$$r = -4, \ -4$$

With equal real roots, the complementary solution is

$$y_c(x) = c_1 e^{-4x} + c_2 x e^{-4x}$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{e^{-4x}, \ xe^{-4x}\}$$

Find the Wronskian for the fundamental solution set.

$$W(e^{-4x}, xe^{-4x}) = \begin{vmatrix} e^{-4x} & xe^{-4x} \\ -4e^{-4x} & e^{-4x} - 4xe^{-4x} \end{vmatrix}$$

$$W(e^{-4x}, xe^{-4x}) = e^{-8x} - 4xe^{-8x} + 4xe^{-8x}$$

$$W(e^{-4x}, xe^{-4x}) = e^{-8x}$$

Then the particular solution is

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)} \, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)} \, dx$$

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$$y_p(x) = -e^{-4x}\int \frac{xe^{-4x}(21x^5e^{-4x})}{e^{-8x}}\,dx + xe^{-4x}\int \frac{e^{-4x}(21x^5e^{-4x})}{e^{-8x}}\,dx$$

$$y_p(x) = -21e^{-4x}\int x^6\,dx + 21xe^{-4x}\int x^5\,dx$$

$$y_p(x) = -3e^{-4x}(x^7) + 7xe^{-4x}\left(\frac{1}{2}x^6\right)$$

$$y_p(x) = \frac{1}{2}x^7e^{-4x}$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1e^{-4x} + c_2xe^{-4x} + \frac{1}{2}x^7e^{-4x}$$

■ 4. Solve the differential equation.

$$y'' + 6y' + 9y = \frac{1}{e^{3x}\sqrt{1-x^2}}$$

Solution:

The associated homogeneous equation is

$$y'' + 6y' + 9y = 0$$

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and solving the characteristic equation gives equal real roots.

$$r^2 + 6r + 9 = 0$$

$$(r + 3)^2 = 0$$

$$r = -3, \ -3$$

With equal real roots, the complementary solution is

$$y_c(x) = c_1 e^{-3x} + c_2 x e^{-3x}$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{e^{-3x}, \ xe^{-3x}\}$$

Find the Wronskian for the fundamental solution set.

$$W(e^{-3x}, xe^{-3x}) = \begin{vmatrix} e^{-3x} & xe^{-3x} \\ -3e^{-3x} & e^{-3x} - 3xe^{-3x} \end{vmatrix}$$

$$W(e^{-3x}, xe^{-3x}) = e^{-6x} - 3xe^{-6x} + 3xe^{-6x}$$

$$W(e^{-3x}, xe^{-3x}) = e^{-6x}$$

Then the particular solution is

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)} \, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)} \, dx$$

$$y_p(x) = -e^{-3x} \int \frac{xe^{-3x}}{e^{-6x} \cdot e^{3x}\sqrt{1 - x^2}} \, dx + xe^{-3x} \int \frac{e^{-3x}}{e^{-6x} \cdot e^{3x}\sqrt{1 - x^2}} \, dx$$

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$$\int \frac{xe^{-3x}}{e^{-6x} \cdot e^{3x}\sqrt{1-x^2}}\,dx = \int \frac{x}{\sqrt{1-x^2}}\,dx = -\sqrt{1-x^2}$$

$$\int \frac{e^{-3x}}{e^{-6x} \cdot e^{3x}\sqrt{1-x^2}}\,dx = \int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x$$

$$y_p(x) = -e^{-3x}(-\sqrt{1-x^2}) + xe^{-3x}\arcsin x$$

$$y_p(x) = e^{-3x}\sqrt{1-x^2} + xe^{-3x}\arcsin x$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1e^{-3x} + c_2xe^{-3x} + e^{-3x}\sqrt{1-x^2} + xe^{-3x}\arcsin x$$

■ 5. Solve the differential equation.

$$y'' + y = \frac{1}{\sin^3 x}$$

Solution:

The associated homogeneous equation is

$$y'' + y = 0$$

and solving the characteristic equation gives complex conjugate roots.

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$$r^2 + 1 = 0$$

$$r = \pm i$$

With complex conjugate roots, the complementary solution is

$$y_c(x) = c_1 \cos x + c_2 \sin x$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{\cos x,\ \sin x\}$$

Find the Wronskian for the fundamental solution set.

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W(\cos x, \sin x) = \cos^2 x + \sin^2 x$$

$$W(\cos x, \sin x) = 1$$

Then the particular solution is

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -\cos x \int \frac{\sin x}{\sin^3 x}\, dx + \sin x \int \frac{\cos x}{\sin^3 x}\, dx$$

$$y_p(x) = -\cos x \int \frac{1}{\sin^2 x}\, dx + \sin x \int \frac{\cos x}{\sin^3 x}\, dx$$

$$\int \frac{1}{\sin^2 x}\, dx = -\cot x$$

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$$\int \frac{\cos x}{\sin^3 x}\, dx = -\frac{1}{2\sin^2 x}$$

$$y_p(x) = -\cos x(-\cot x) + \sin x\left(-\frac{1}{2\sin^2 x}\right)$$

$$y_p(x) = \frac{\cos(2x)}{2\sin x}$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1\cos x + c_2\sin x + \frac{\cos(2x)}{2\sin x}$$

6. Solve the differential equation.

$$y'' + 4y = \cos^2(2x)$$

Solution:

The associated homogeneous equation is

$$y'' + 4y = 0$$

and solving the characteristic equation gives complex conjugate roots.

$$r^2 + 4 = 0$$

$$r = \pm 2i$$

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With complex conjugate roots, the complementary solution is

$$y_c(x) = c_1 \cos(2x) + c_2 \sin(2x)$$

and the fundamental set of solutions is

$$\{y_1, y_2\} = \{\cos(2x),\ \sin(2x)\}$$

Find the Wronskian for the fundamental solution set.

$$W(\cos(2x), \sin(2x)) = \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix}$$

$$W(\cos(2x), \sin(2x)) = 2\sin^2(2x) + 2\cos^2(2x)$$

$$W(\cos(2x), \sin(2x)) = 2$$

Then the particular solution is

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -\cos(2x) \int \frac{\sin(2x)\cos^2(2x)}{2}\, dx + \sin(2x) \int \frac{\cos(2x)(1 - \sin^2(2x))}{2}\, dx$$

$$\int \frac{\sin(2x)\cos^2(2x)}{2}\, dx = -\frac{\cos^3(2x)}{12}$$

$$\int \frac{\cos(2x)(1 - \sin^2(2x))}{2}\, dx = \frac{\sin(2x)}{4} - \frac{\sin^3(2x)}{12}$$

$$y_p(x) = -\cos(2x)\left(-\frac{\cos^3(2x)}{12}\right) + \sin(2x)\left(\frac{\sin(2x)}{4} - \frac{\sin^3(2x)}{12}\right)$$

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$$y_p(x) = \frac{\sin^2(2x)}{4} + \frac{\cos(4x)}{12}$$

Then the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 \cos(2x) + c_2 \sin(2x) + \frac{\sin^2(2x)}{4} + \frac{\cos(4x)}{12}$$

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INITIAL VALUE PROBLEMS WITH NONHOMOGENEOUS EQUATIONS

■ 1. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 2$.

$$y'' - 2y' - 3y = 10e^x \sin x$$

Solution:

Solving the characteristic equation for the associated homogeneous equation gives

$$y'' - 2y' - 3y = 0$$

$$r^2 - 2r - 3 = 0$$

$$(r + 1)(r - 3) = 0$$

$$r = -1, 3$$

so the complementary solution with distinct real roots is

$$y_c(x) = c_1 e^{-x} + c_2 e^{3x}$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = e^x(A \sin x + B \cos x)$$

$$y_p'(x) = e^x((A - B)\sin x + (A + B)\cos x)$$

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$$y_p''(x) = e^x(-2B\sin x + 2A\cos x)$$

Plugging these into the original differential equation, we get

$$e^x(-2B\sin x + 2A\cos x) - 2e^x((A-B)\sin x + (A+B)\cos x)$$

$$-3e^x(A\sin x + B\cos x) = 10e^x\sin x$$

$$e^x(-5A\sin x - 5B\cos x) = 10e^x\sin x$$

Using undetermined coefficients, we find $A = -2$ and $B = 0$, so the particular solution is

$$y_p(x) = -2e^x\sin x$$

and the general solution and its derivative are

$$y(x) = c_1e^{-x} + c_2e^{3x} - 2e^x\sin x$$

$$y'(x) = -c_1e^{-x} + 3c_2e^{3x} - 2e^x\sin x - 2e^x\cos x$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 2$.

$$c_1 + c_2 - 0 = 0$$

$$-c_1 + 3c_2 - 0 - 2 = 2$$

We'll simplify these equations to get

$$c_1 = -c_2$$

$$c_2 + 3c_2 = 4$$

With $c_1 = -1$ and $c_2 = 1$, the solution to the initial value problems is

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$$y(x) = -e^{-x} + e^{3x} - 2e^{x} \sin x$$

■ 2. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' - 4y = 8x^2 + 4$$

Solution:

Solving the characteristic equation for the associated homogeneous equation gives

$$y'' - 4y = 0$$

$$r^2 - 4 = 0$$

$$r = -2, 2$$

so the complementary solution with distinct real roots is

$$y_c(x) = c_1 e^{2x} + c_2 e^{-2x}$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = Ax^2 + Bx + C$$

$$y_p'(x) = 2Ax + B$$

$$y_p''(x) = 2A$$

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Plugging these into the original differential equation, we get

$$2A - 4(Ax^2 + Bx + C) = 8x^2 + 4$$

$$2A - 4Ax^2 + 4Bx + 4C = 8x^2 + 4$$

Using undetermined coefficients, we find $A = -2$, $B = 0$, and $C = -2$, so the particular solution is

$$y_p(x) = -2x^2 - 2$$

and the general solution and its derivative are

$$y(x) = c_1 e^{2x} + c_2 e^{-2x} - 2x^2 - 2$$

$$y'(x) = 2c_1 e^{2x} - 2c_2 e^{-2x} - 4x$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$.

$$c_1 + c_2 - 0 - 2 = 0$$

$$2c_1 - 2c_2 - 0 = 0$$

We'll simplify these equations to get

$$c_1 + c_2 = 2$$

$$c_1 = c_2$$

With $c_1 = 1$ and $c_2 = 1$, the solution to the initial value problems is

$$y(x) = e^{2x} + e^{-2x} - 2x^2 - 2$$

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■ 3. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' - 10y' + 25y = 28x^6e^{5x}$$

Solution:

Solving the characteristic equation for the associated homogeneous equation gives

$$y'' - 10y' + 25y = 0$$

$$r^2 - 10r + 25 = 0$$

$$(r - 5)^2 = 0$$

$$r = 5, 5$$

so the complementary solution with equal real roots is

$$y_c(x) = c_1e^{5x} + c_2xe^{5x}$$

The fundamental set of solutions is

$$\{y_1, y_2\} = \{e^{5x}, xe^{5x}\}$$

and the Wronskian for the fundamental set is

$$W(e^{5x}, xe^{5x}) = \begin{vmatrix} e^{5x} & xe^{5x} \\ 5e^{5x} & e^{5x} + 5xe^{5x} \end{vmatrix}$$

$$W(e^{5x}, xe^{5x}) = e^{10x} + 5xe^{10x} - 5xe^{10x}$$

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$$W(e^{5x}, xe^{5x}) = e^{10x}$$

Then the particular solution is

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

$$y_p(x) = -e^{5x} \int \frac{xe^{5x}(28x^6 e^{5x})}{e^{10x}}\, dx + xe^{5x} \int \frac{e^{5x}(28x^6 e^{5x})}{e^{10x}}\, dx$$

$$y_p(x) = -e^{5x} \int 28x^7\, dx + xe^{5x} \int 28x^6\, dx$$

$$y_p(x) = -e^{5x} \left(\frac{7}{2}x^8\right) + xe^{5x}(4x^7)$$

$$y_p(x) = \frac{1}{2}x^8 e^{5x}$$

The general solution and its derivative are

$$y(x) = c_1 e^{5x} + c_2 xe^{5x} + \frac{1}{2}x^8 e^{5x}$$

$$y'(x) = 5c_1 e^{5x} + c_2 e^{5x} + 5c_2 xe^{5x} + 4x^7 e^{5x} + \frac{5}{2}x^8 e^{5x}$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 0$.

$$c_1 + 0 + 0 = 1$$

$$5c_1 + c_2 + 0 + 0 + 0 = 0$$

We'll simplify these equations to get

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$$c_1 = 1$$

$$c_2 = -5c_1 = -5$$

With $c_1 = 1$ and $c_2 = -5$, the solution to the initial value problems is

$$y(x) = e^{5x} - 5xe^{5x} + \frac{1}{2}x^8 e^{5x}$$

■ 4. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 1$ and $y'(0) = 0$.

$$4y'' + 4y' + y = 25\sin x + 4$$

Solution:

Solving the characteristic equation for the associated homogeneous equation gives

$$4y'' + 4y' + y = 0$$

$$4r^2 + 4r + 1 = 0$$

$$(2r + 1)^2 = 0$$

$$r = -\frac{1}{2}, \ -\frac{1}{2}$$

so the complementary solution with equal real roots is

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$$y_c(x) = c_1 e^{-\frac{1}{2}x} + c_2 x e^{-\frac{1}{2}x}$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = A\sin x + B\cos x + C$$

$$y_p'(x) = A\cos x - B\sin x$$

$$y_p''(x) = -A\sin x - B\cos x$$

Plugging these into the original differential equation, we get

$$4(-A\sin x - B\cos x) + 4(A\cos x - B\sin x)$$

$$+A\sin x + B\cos x + C = 25\sin x + 4$$

$$(-3A - 4B)\sin x + (4A - 3B)\cos x + C = 25\sin x + 4$$

Using undetermined coefficients, we find $A = -3$, $B = -4$, and $C = 4$, so the particular solution is

$$y_p(x) = -3\sin x - 4\cos x + 4$$

and the general solution and its derivative are

$$y(x) = c_1 e^{-\frac{1}{2}x} + c_2 x e^{-\frac{1}{2}x} - 3\sin x - 4\cos x + 4$$

$$y'(x) = -\frac{1}{2}c_1 e^{-\frac{1}{2}x} + c_2 e^{-\frac{1}{2}x} - \frac{1}{2}c_2 x e^{-\frac{1}{2}x} - 3\cos x + 4\sin x$$

Substitute the initial conditions $y(0) = 1$ and $y'(0) = 0$.

$$c_1 + 0 - 4 + 4 = 1$$

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$$-\frac{1}{2}c_1 + c_2 + 0 - 3 + 0 = 0$$

We'll simplify these equations to get

$$c_1 = 1$$

$$c_2 = \frac{7}{2}$$

With $c_1 = -1$ and $c_2 = 7/2$, the solution to the initial value problems is

$$y(x) = e^{-\frac{1}{2}x} + \frac{7}{2}xe^{-\frac{1}{2}x} - 3\sin x - 4\cos x + 4$$

■ 5. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' + 4y' + 5y = 17e^{2x} + 40\sin(3x)$$

Solution:

Solving the characteristic equation for the associated homogeneous equation gives

$$y'' + 4y' + 5y = 0$$

$$r^2 + 4r + 5 = 0$$

$$(r+2)^2 + 1 = 0$$

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$$r = -2 \pm i$$

so the complementary solution with complex conjugate roots is

$$y_c(x) = c_1 e^{-2x} \cos x + c_2 e^{-2x} \sin x$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = Ae^{2x} + B\sin(3x) + C\cos(3x)$$

$$y_p'(x) = 2Ae^{2x} + 3B\cos(3x) - 3C\sin(3x)$$

$$y_p''(x) = 4Ae^{2x} - 9B\sin(3x) - 9C\cos(3x)$$

Plugging these into the original differential equation, we get

$$4Ae^{2x} - 9B\sin(3x) - 9C\cos(3x) + 4(2Ae^{2x} + 3B\cos(3x) - 3C\sin(3x))$$

$$+5(e^{2x} + B\sin(3x) + C\cos(3x)) = 17e^{2x} + 40\sin(3x)$$

$$17Ae^{2x} + (-4B - 12C)\sin(3x) + (12B - 4C)\cos(3x) = 17e^{2x} + 40\sin(3x)$$

Using undetermined coefficients, we find $A = 1$, $B = -1$, and $C = -3$, so the particular solution is

$$y_p(x) = e^{2x} - \sin(3x) - 3\cos(3x)$$

and the general solution and its derivative are

$$y(x) = c_1 e^{-2x}\cos(x) + c_2 e^{-2x}\sin(x) + e^{2x} - \sin(3x) - 3\cos(3x)$$

$$y'(x) = -2e^{-2x}(c_1\cos(x) + c_2\sin(x)) + e^{-2x}(-c_1\sin(x) + c_2\cos(x))$$

$$+2e^{2x} - 3\cos(3x) + 9\sin(3x)$$

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Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$.

$$c_1 + 0 + 1 + 0 - 3 = 0$$

$$-2c_1 + c_2 + 2 - 3 = 1$$

We'll simplify these equations to get

$$c_1 = 2$$

$$c_2 = 6$$

With $c_1 = 2$ and $c_2 = 6$, the solution to the initial value problems is

$$y(x) = 2e^{-2x}\cos(x) + 6e^{-2x}\sin(x) + e^{2x} - \sin(3x) - 3\cos(3x)$$

■ 6. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 2$ and $y'(0) = 2$.

$$y'' + 16y = 40\sin^2(3x)$$

Solution:

Solving the characteristic equation for the associated homogeneous equation gives

$$y'' + 16y = 0$$

$$r^2 + 16 = 0$$

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$$r = \pm 4i$$

so the complementary solution with complex conjugate roots is

$$y_c(x) = c_1 \cos(4x) + c_2 \sin(4x)$$

Our guess for the particular solution and its derivatives will be

$$y_p(x) = A\cos(6x) + B\sin(6x) + C$$

$$y_p'(x) = -6A\sin(6x) + 6B\cos(6x)$$

$$y_p''(x) = -36A\cos(6x) - 36B\sin(6x)$$

Plugging these into the original differential equation, we get

$$-36A\cos(6x) - 36B\sin(6x) + 16(A\cos(6x) + B\sin(6x) + C) = 40\sin^2(3x)$$

Using undetermined coefficients, we find $A = 1$, $B = 0$, and $C = 5/4$, so the particular solution is

$$y_p(x) = \cos(6x) + \frac{5}{4}$$

and the general solution and its derivative are

$$y(x) = c_1 \cos(4x) + c_2 \sin(4x) + \cos(6x) + \frac{5}{4}$$

$$y'(x) = -4c_1 \sin(4x) + 4c_2 \cos(4x) - 6\sin(6x)$$

Substitute the initial conditions $y(0) = 2$ and $y'(0) = 2$.

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$$c_1 + 0 + 1 + \frac{5}{4} = 2$$

$$0 + 4c_2 - 0 = 2$$

We'll simplify these equations to get

$$c_1 = -\frac{1}{4}$$

$$c_2 = \frac{1}{2}$$

With $c_1 = -1/4$ and $c_2 = 1/2$, the solution to the initial value problems is

$$y(x) = -\frac{1}{4}\cos(4x) + \frac{1}{2}\sin(4x) + \cos(6x) + \frac{5}{4}$$

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